

0/1 Knapsack Problem:

The 0/1 Knapsack problem is an optimization problem where we have a bag with a fixed capacity and several items with different weights and profits.

Algorithm:

Step 1: Start

Step 2: Read the number of items n .

Step 3: Read the weight and profit of each item.

Step 4: Read the knapsack capacity W .

Step 5: Create a DP table $dp[n+1][W+1]$.

Step 6: For each item i and capacity w :

- If the item's weight is less than or equal to w :

$dp[i][w] = \max(\text{profit}[i-1] + dp[i-1][w-\text{weight}[i-1]], dp[i-1][w])$

- Otherwise:

$dp[i][w] = dp[i-1][w]$

Step 7: $dp[n][W]$ gives the maximum profit.

Step 8: Stop.

Out put:

```
Enter weight of item 3: 3
Enter profit of item 3: 30
Enter knapsack capacity: 4
Maximum Profit = 40

...Program finished with exit code 0
Press ENTER to exit console.
```

Time Complexity

$O(n \times W)$

Where:

- n = number of items
- W = knapsack capacity

Space Complexity

$O(n \times W)$

Because we create a DP table of size $(n+1) \times (W+1)$.